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MILITARY SPECCIFICATION

CIRCUIT CARD ASSEMBLY, SIGHT CURRENT GENERATOR

GENERAL SPECIFICATION FOR

This specification is approved for use withtn the U.S. Army Armament. Munitions and Chemical Command, and is available for use by ail Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scoDe. This specification covers the general requirements for circuit card assemblies which form a part of the Sight Current Generator (see 6.1).

2* APPLICABLE D O C U M E N T S

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of SpecIfications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 Fire Control *Materiel*, Manufacture and Inspection, General Specification for .

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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MIL-X 5607 Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109 Quality Assurance Terms and Definitions
MIL-STD-810 Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, Standards, and handbooks are available from the standardization Documents order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS see 6.4)

US. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10545465 Isolator
12000239 Sight current Generator

(Copies of drawings publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on specific circuit card assembly drawing, all associated -drawings, and with all requirements specified in applicable specifications and standards. where applicable, the circuit card assemblies shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. INORGANic protective surface finishings"
Part identification and marking
c Workmanship

3.3 Environmental service conditions.

3.3.1 vibration. The circuit card assemblies when mounted in a normal service condition (or equivalent) shall not suffer damage and shall meet the requirements of 3.4 and 3.s after exposure to sinusoidal vibration, in each of three mutually perpendicular axis, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

-r The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decode.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

NOrmal service condition is defined by Sight Current Generator Drawing 12000239 and Isolator Drawing 10545465.

3.3.2 Shock. The circuit card assemblies when mounted in a normal servic condition (or equivalent) shall not suffer damage and shall;meet the requirements of" 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10% , 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

Normal service condition is defined by Sight Current Generator Drawing. 120002%9 and Islator Drawing M545465i

3.3.3 Altitude.

3.3.3.1 Operating. The circuit card assemblies shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of HG.).

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3.3.3.2 Non-operating. The circuit card assemblies shall meet the requirements of 3.4 and 3.S after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg}.

3.3.4 Temperature.

3.3.4.1 Operating. The circuit card assemblies shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 Degrees F and +155 Degrees F.

3.3.4.2 Non-operatlng (storage). The circuit card assemblies shall not suffer permanent damage when subjected to the constant temperature extremes of -60 Degrees F for a period of 24 hours and +160 Degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The circuit card assemblies shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a-relative humidity in excess of 95-per-cent for a period of 10 days.

3.4 Electrical. As specified on the applicable circuit card assembly detail specification.

3.5 Performance . As specified on the applicable circuit card assembly detail specification.

3.6 **First article.** When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein {sea 4.3).

3.7 Workmanship. The workmanship requirements of MIL-F-13926 shall apply.-. .

3.8 Soldering. AU solderimg shall be in accordance-with the genezal requirements of MIL-STD-2C)00 in addition to detail requirement task 1' of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspecti- tlnl~"f otherwt=e specif~ed lm the contract or purchase order, the contractor 1s responsible for" the performance of all inspection requirements (examinatibas and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor~s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responskbility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission *of known* defective material, either indtcated *or* actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-139'26 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection-{see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinatzons and tests shall--be-conducted at the following-standard ambient conditions:

a.	Temperature	+5Q deg to +9o deg °F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of *Hg*.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall- submit a first article sample as designated '~y the contracting offlcef'" "... consisting of three circuit card assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test Specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I (fictional Test) and Table II (Special Sailpling) herein and to *any* or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

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4.4 **Quality** conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection **of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturs, in one unchanged process, using the same materials.and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.**

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies subassemblies shall inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data package (TDP). In the absence of, applicable quality assurance provisions of MIL-F-13926 shall apply.-- .

4.4.2.2 Final acceptance inspection. subsequent to first article approval, examination tests related.to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests **in Table I shall be performed** on a 100% basis.

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TABLE 1. Functional tests
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4*7.3
Minor:	None	N/A

4.5 Special sampling. One circuit card assembly out-of the first sO produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirments and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract the Items shall be returned to the lot upon successful completion of testing.

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accompllsh~ in accordance with -- the provisions of the contract.

TABLE 11. Special sampling

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103 vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105 . Altitude	3.3.3	4.7.2.3
106. temperature	3 . 3 . 4	4.7.1.4
107 . Humidity	3.3.5	4 . 7 . 1 . 5
Minor:	None	N/A

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, SUpply and maintenance of test equipment shall be in accordance wtih the applicable requirements of MIL-I-45607.

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the 'Methods of Inspection specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and 'Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, Construction, materials, dimensions, and tolerances used in the design of test 'equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed go percent of the prescribed tolerance limit and permit positive rejection when nonconforming. construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equip ment designs shall be submitted for approval in accordance with 6.3. -

4.6.2.3 Test equipment. As specified in the applicable circuit card assembly detail specification.

4.7 Methods of inspection.

4.7.1 Environmental. environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2. 1 apply. The circuit card assemblies shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the circuit card assemblies shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of procedure 1) need not be conducted until completion of all shock Impulses.

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b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The **maximum g shall occur at 5.5 milliseconds ± 10%.**

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the circuit card assemblies shall be tested as specified in 4.7.2 and 4.7.3 to determine **compliance with 3.3.2.**

4.7.1.3 Altitude (operating and non-operating) This test shall be performed *as* specified in Procedures II and 111, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method **501 of** MIL-STD-810 with constant temperature extremes as specified in 3*3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 *LOW (*operating and non-operating) This test shall be performed as specified in procedures X-and XI; -Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and *4.7.3* for compliance with 3.3.4.

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4.7.1.5 **Humidity (non-operating. This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except** that in Step 7, checkout shall be performed at the conclusion of-the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 **Electrical. AS specified in the applicable circuit card assembly detail specification.**

4.7.3 Performance. As specified in the applicable circuit card assembly detail specification.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-ZOOO as specified in paragraph 3.7.

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5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with packaging Data Sheet specified on applicable circuit card assembly detail specification. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cIRcuit card assemblies covered by this speci,fication are plug in type components intended for use in the Sight Current Generator.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title., number and date" of this specificati,On and applicable detail spec i.fication.
b. Applicable stock number.
. First Article Sample requirements.
c. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Circuit card assembly part number.
9. Certificate of conformance for each lot or shipment of

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e **6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue** due to the extensiveness of the changes.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-50785A (AR)	15 May 1990

3. DOCUMENT TITLE

CIRCUIT CARD ASSEMBLY, SIGHT CURRENT GENERATOR GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME US Army ARDEC Standardization & Specification Office	b. TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6674 (2) AUTOVON: 880-6674	
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